WEEK 7:
HUMAN-IN-THE-LOOP AI
& AI AUDIT

COMP0195

Shreevanth K. Gopalakrishnan

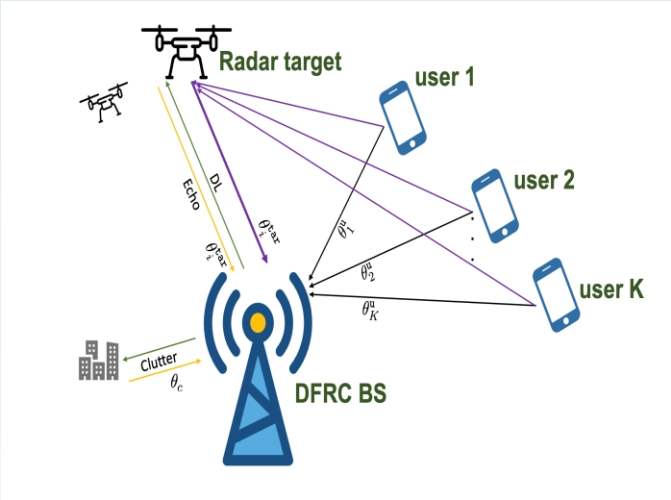
Centre for Doctoral Training in Cybersecurity

March 2025

Speaker Background

Now:

- 2nd year PhD student in Cybersecurity: privacy concerns of wireless sensing to track human movements and presence.
- Under the umbrella of Integrated Sensing and Communications & 6G.



[1]

Previously:

- Previous research on the security of Industrial IoT and BACS.
- MSc Applied AI. Thesis on human factors and AI in aerospace.
- MEng Aerospace Engineering. Thesis on 3D-printable Hydrogen fuel cells.

[2]

Intended Learning Outcomes

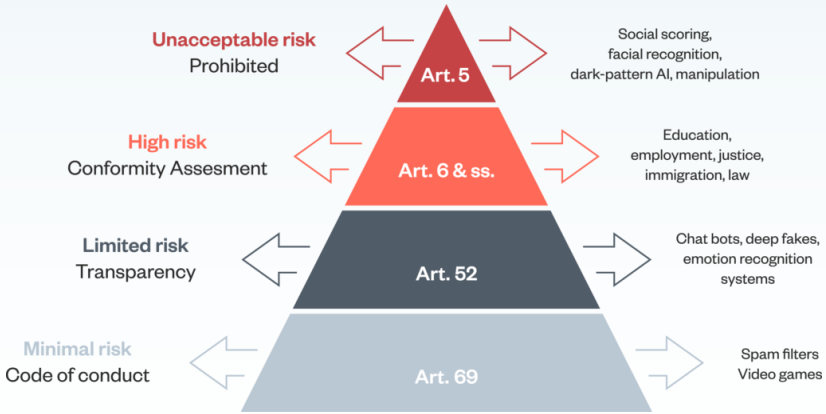
Part 1: Human-in-the-loop AI/ML

- *What:* When and where could/should humans be a part of that?
- *Why:* We want to keep our jobs; contribute to improved model and human performance/efficiency.

[3]

Part 2: AI Audit

- *What:* Independent system/algorithm risk-based evaluation w.r.t. an accepted standard.
- *Why:* AI Proliferation must not have an adverse effect on our societal values and institutions.

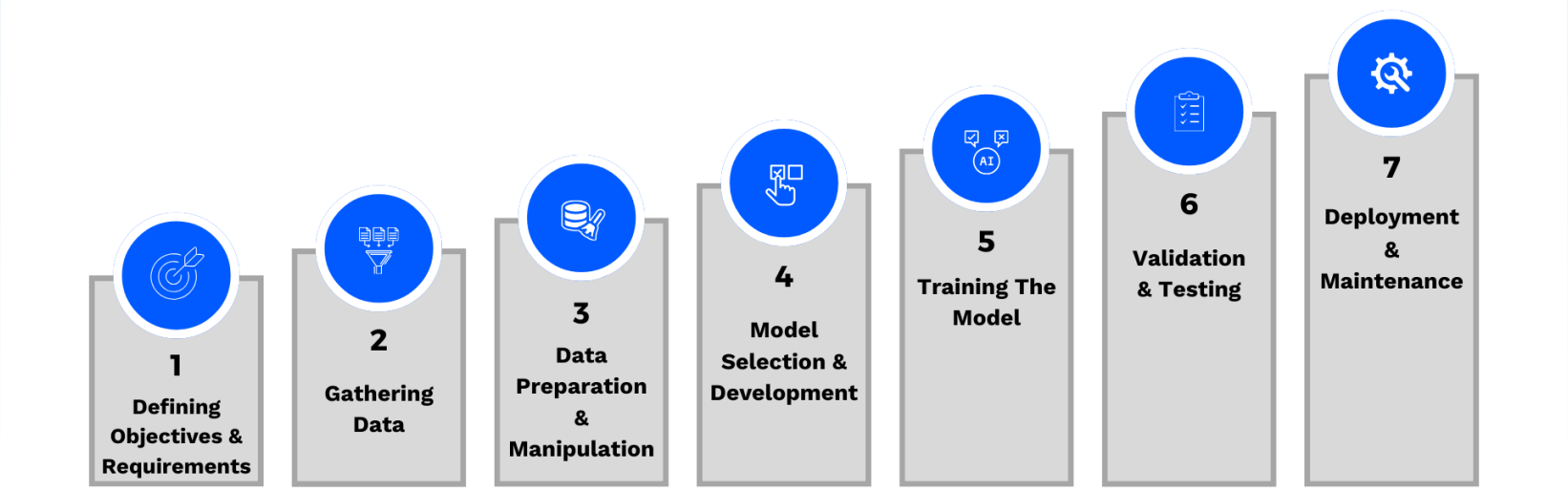


[4]

PART 1:
HUMAN-IN-THE-LOOP AI/ML

Process of Developing AI

- Humans are required at various points in the loop of the machine learning process: data collection, modelling, model development, testing, etc.
- Once the software is released to the public/to customers, can it be updated if required?
- Key reference for HITL-AI: Mosqueira-Rey et al. 2023 [5]



[6]

Software Development Paradigms for AI

- **Monolithic**: single, large, unified unit, where all functionalities are tightly coupled within one codebase.

- Alternatives:
  - ❑ **Microservices approach:** Different functionalities are implemented as independent, smaller modules developed, deployed, and updated separately, promoting greater agility and flexibility.
  - ❑ **Specialized models:** Specific tasks are catered to by separate models, leading to better performance and adaptability.

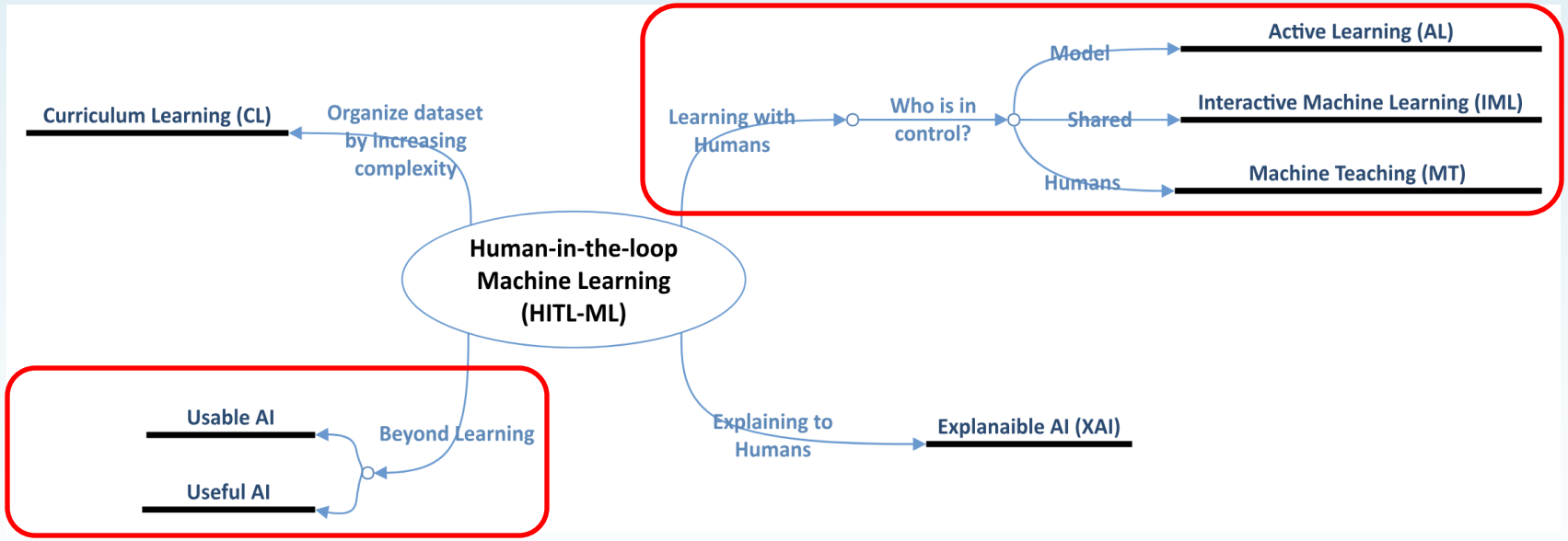
- Naturally, there are some **trade-offs**. E.g., larger, more complex model/organisation prefer a modular approach.

Motivation for Human-In-the-Loop-ML

- We are interested in how humans interact with the ML models. This falls under the broad field of **Human-in-the-loop machine learning (HITL-ML)** [7].

- Why should we investigate how/whether humans and ML algorithms should work together? Because there is potential:
  - ❑ To make **machine learning** more powerful.
  - ❑ And also to make **humans** more effective and more efficient.

Categories of HITL-ML



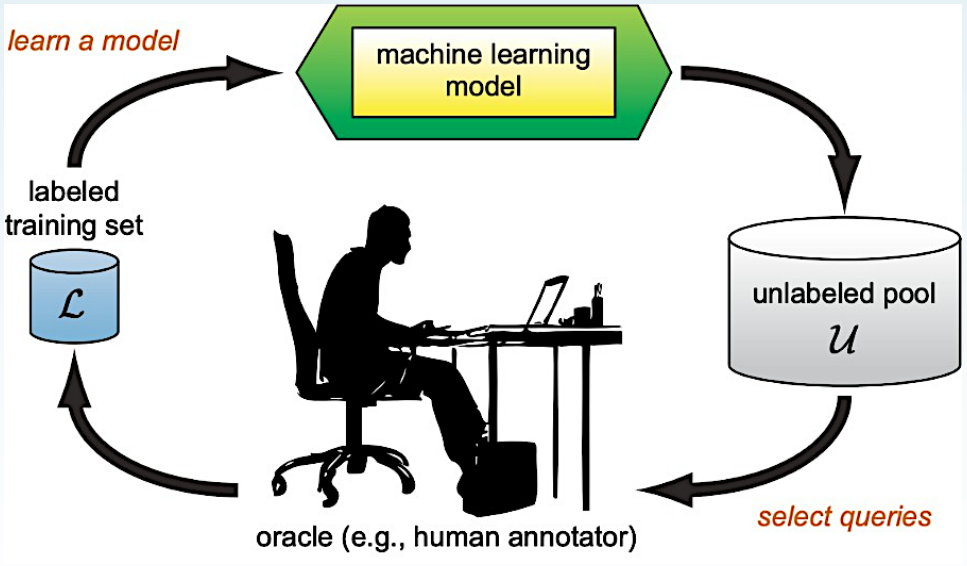
[5]

Active Learning (AL)

- The **system remains in control** and uses humans as oracles to annotate data.

- **The learner** queries an entity with extensive knowledge of the domain for annotating unlabelled examples.

- Effective when there is a lot of unlabelled and complex data which **humans can annotate efficiently**.



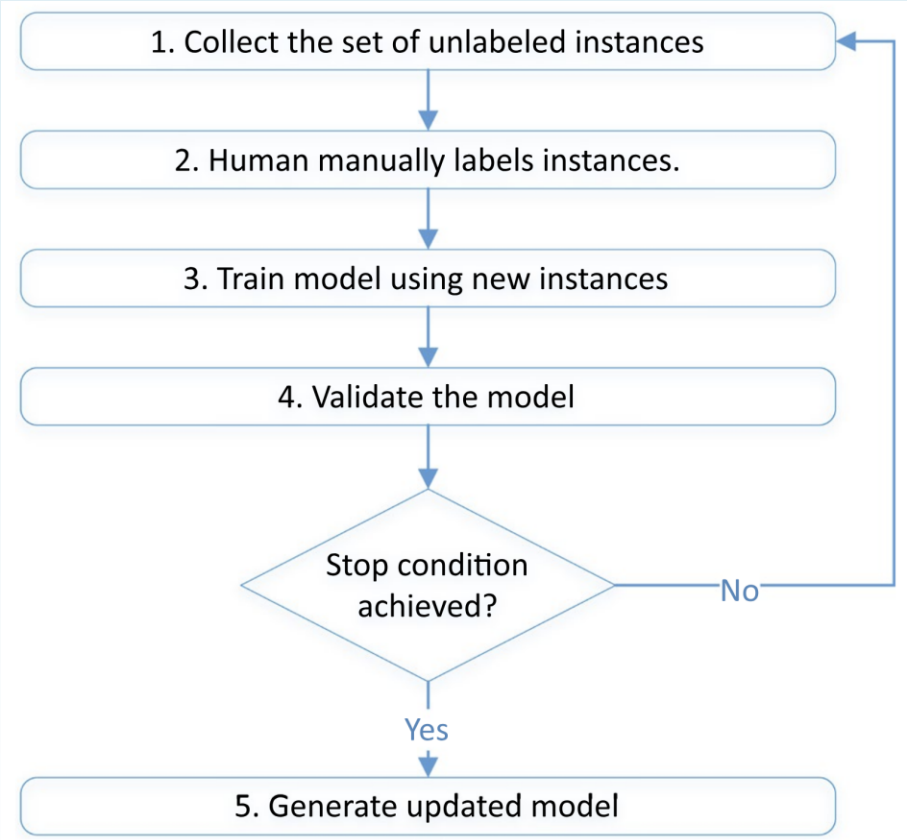
[8]

Active Learning Process

- Data set is divided into two groups:
  - ❑ **labelled and unlabelled**.

- The model iteratively **selects new unlabelled** example(s) and provides it to the oracle **for labelling**.

- Then **trains the model** using the new data until a stop condition is achieved.



[5]

Applications of AL

- The cost of **annotating data** is high, but it is **a task humans are adept at**, e.g., interpreting images or processing language.
- The active learner attempts to maximise their performance/accuracy **using as few labelled instances** as possible.

- It is **quite mature** and is widely used in the medical field:
  - ❑ Classification of radiology reports [9],[10]
  - ❑ Recognizing multiple types of plankton [11]
  - ❑ Serving as the basis for clinical decision-making [12]
  - ❑ Classifying cancer pathology reports [13]

AL Limitations

- **Noisy oracles**: how to decide an oracle label is trustworthy
- **Variable labelling cost**: The cost of obtaining new labels has been assumed uniform and fixed
- **Changing model classes**: In some scenarios the model does not contain a representative set of examples of a real problem

...

Interactive Machine Learning (IML)

- The fundamental parts of an IML system as: **users, model, data and interface**.
- The **interface** part is responsible for the **bidirectional feedback** between the other three components and for the authors [14].
- Key features of IML:
  - ❑ **Humans in the ML loop**: assigned to tasks at which they are more efficient than machines.
  - ❑ **Humans assuming different roles**: can be domain experts, non-expert users, data scientists, etc.
  - ❑ **Incremental methodology**: model is updated iteratively and incrementally.
  - ❑ **Importance of the user interface**: influences how learning takes place and conditions learning outcomes.

Interactive Machine Learning Process

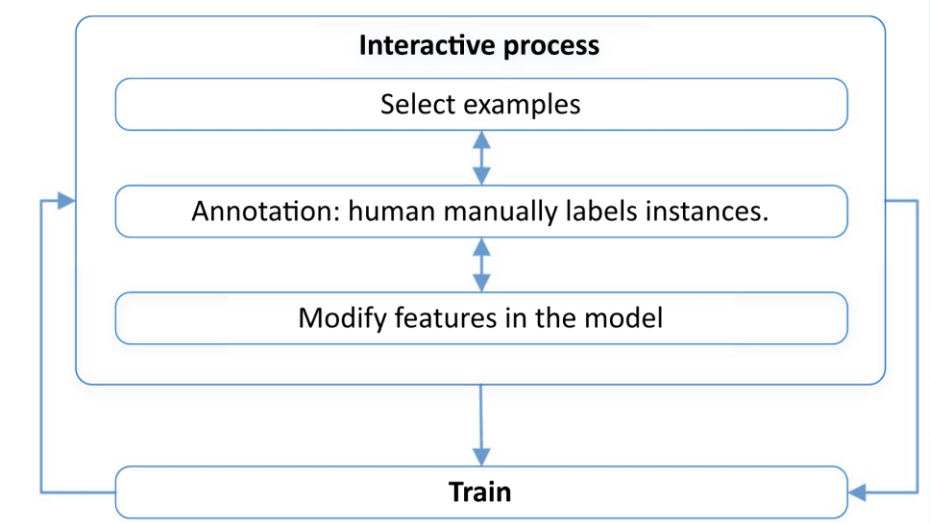
Based on AL, but more freedom:

- **AL is the basis** for IML.

- In AL the model retains the control and uses the human as an oracle; **in IML the control is shared.**

- **Human-Computer Interaction** techniques (HCI) is more important in IML.

- In IML, **humans perform more tasks other than labelling data** in a freer and less structured process.



[5]

Applications of IML

Image and Video:

- Users create custom rules for image classification [15].
- Users mark relevant content on images/videos for interactive segmentation [16], [17].

Other Timeseries:

- Music composition suited to user's technique, environment, goals: [18]; interactive sound recognition: [19].
- Development of brain-computer interfaces [20].

Text:

- Facilitating citation screening for systematic reviews [21].

Complex Data (large datasets and/or messy/containing hidden structure):

- Telemedicine for automatic & personalised interventions [22].

IML Limitations

Similar to AL, but adds some of its own:

- ML aspects to be **intermingled with HCI aspects**.

- Another drawback that arises in IML and that it also shares with other interactive models is the **dependence on the presence of human experts**.

- Must deal with **limitations of humans themselves** such as availability, attention, interactivity, different expertise, etc.

Machine Teaching (MT)

- Places **responsibility firmly on the teacher**.

- In recent times, Machine Teaching is used to describe **a teacher who teaches an ML model to an ML algorithm**.

- The teacher can be a **human or an algorithm itself**.

- **Different to machines teaching humans**, e.g., Intelligent Computer-Aided Instruction or Intelligent Tutoring Systems.

- N.b., the terminologies used can be quite confusing.

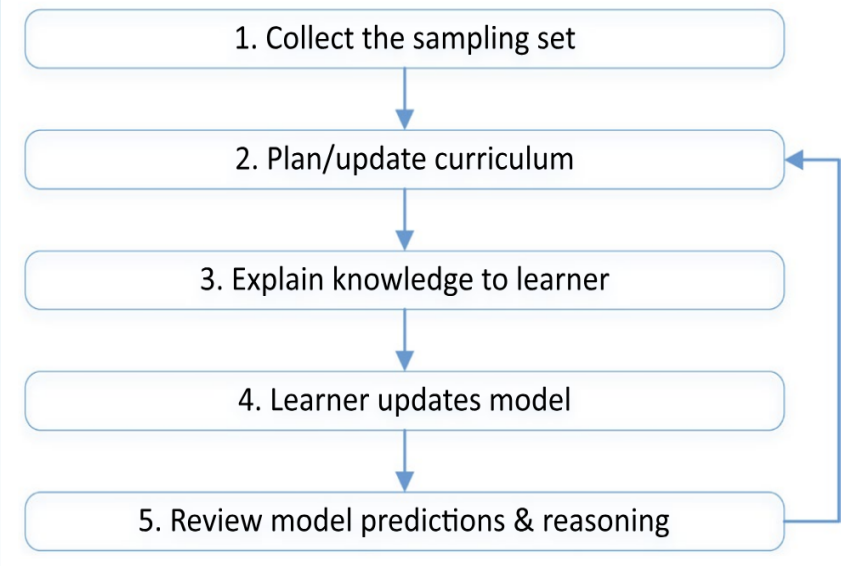
Humans as Teachers or *Interactive* MT

- Enables us to **leverage intrinsic human capabilities related to teaching**.

- **Makes building ML models more accessible** to non-ML experts, e.g., AutoML (however, this has issues with transparency).

- Facilitates the **creation of semantic and debuggable models**.

- Here, **human expert prepares curriculum, explains knowledge, then assesses model predictions & reasoning**.



[5]

Machines as Teachers or *Iterative* MT

- Here, **the teacher (a machine itself) knows the target model in advance and also the student's algorithm, and must "teach" the former to the latter somehow**.

- Based on the level of access the teacher has to the learner, teacher implementations can be categorised into:
  - ❑ **Omniscient**: Total access to learner. Strategically feeds examples.
  - ❑ **Surrogate**: Access to loss function. Alternative usefulness calc.
  - ❑ **Imitation**: Uses copy of learner as no direct access.
  - ❑ **Active**: Teacher can make queries to student to monitor status.
  - ❑ **Adaptive**: Observes learner's hypothesis at each iteration.

Applications of MT

- For interactive MT: make human-machine knowledge transfer more accessible, make humans better teachers, making learner more human-like.
- For iterative MT: there must a need to convey a target model/influence a learner via training data.

- **Humans as teachers**:
  - ❑ Teach robot manipulators how to move [23, 24].
  - ❑ Making task-oriented dialog systems more convincing [25].
- **Machines as teachers**:
  - ❑ Training set poisoning, e.g., evading spam filters [26].
  - ❑ Determining the Minimum Viable Data (MVD) [27].

MT Limitations

- Assuming that there is a valid case for using MT, especially iterative MT, **there could still be some limitations**.
- **Teaching with *imperfect* knowledge** of:
  - ❑ Learner
  - ❑ Task specification
- This would result in a **sub-optimal curriculum**.

- Devidze et al. (2020) [28] stated that it is **far more important for a teacher to be able to estimate the learner's learning rate**, compared to what they know

- Deeper question: **how can humans effectively teach models that/to surpass human capabilities**?

Usable AI vs Useful AI

Two new terms related to the relationship between humans and AI models that **go beyond cooperation in learning**, called "usable AI" and "useful AI", and which are fundamental to ensure that an AI model is successful [29]:

- **Usable AI** is an AI solution that is easy to learn and use via optimal user experience (UX) created by effective Human-Computer Interaction (HCI) design

- **Useful AI** is an AI solution that can provide the functions required to satisfy target users' needs (Human-centred AI) in the valid usage scenarios of their work and life. Leads to a discussion about trust.

Trustworthy AI

- Trust is a critical concept in system design because an **imperfect AI is likely to be rejected** unless a reasonable level of trust is generated between humans and the system. When trust is not generated, **people do not accept decisions made by the system**.
- Properties of a trustworthy AI [30]:
  - ❑ Accuracy
  - ❑ Robustness
  - ❑ Fairness
  - ❑ Accountability
  - ❑ Transparency
  - ❑ Explainability
  - ❑ Ethical

Scenario: How are LLM’s Trained?

Scenario: How are LLM's Trained?

Data Collection and Pre-processing → Pre-training: Self-Supervised Learning → Supervised Fine-Tuning → Reinforcement Learning with Human Feedback → Post-Deployment Feedback and Continuous Learning

Pre-training: Self-Supervised Learning	Supervised Fine-Tuning	Reinforcement Learning with Human Feedback	Post-Deployment Feedback and Continuous Learning
No direct human involvement. Compute Heavy Creates **Base Model**	Similar to Machine Teaching Humans prepare data (examples of correct outputs) Creates **Assistant Model**	Direct HITL process Humans rank multiple model responses Feeds into a Reward Model LLM is fine-tuned with RL (PPO)	Some human involvement. Go back to Fine-tuning…

[31]

References for Part 1

[1] https://wireless.engineering.nyu.edu/integrated-sensing-and-communication-isac/
[2] https://www.iotworldtoday.com/metaverse/ptc-bae-team-to-create-augmented-reality-fighter-pilot-helmet-
[3] https://levity.ai/blog/human-in-the-loop
[4] https://www.adalovelaceinstitute.org/resource/eu-ai-act-explainer/
[5] https://doi.org/10.1007/s10462-022-10246-w
[6] https://research.aimultiple.com/developing-ai/
[7] https://www.manning.com/books/human-in-the-loop-machine-learning
[8] http://digital.library.wisc.edu/1793/60660
[9] https://doi.org/10.1145/1143844.1143897
[10] https://doi.org/10.1136/amiajnl-2013-002516
[11] https://doi.org/10.1109/ICPR.2004.1334570
[12] https://doi.org/10.1016/j.media.2021.102062
[13] https://doi.org/10.1186/s12859-021-04047-1
[14] https://doi.org/10.1145/3185517
[15] https://doi.org/10.1145/1357054.1357061
[16] https://doi.org/10.1109/MCSE.2013.74
[17] https://doi.org/10.1038/nmeth.2281
[18] https://doi.org/10.1145/2598153.2598189
[19] https://doi.org/10.1145/3377325.3377483
[20] https://doi.org/10.1145/2723162
[21] https://doi.org/10.1145/2110363.2110464
[22] http://arxiv.org/abs/1803.01842
[23] https://doi.org/10.1177/0278364919884623
[24] https://doi.org/10.1109/ICRA.2018.8461194
[25] https://doi.org/10.1162/tacl_a_00399
[26] http://arxiv.org/abs/1801.05927
[27] https://doi.org/10.1016/j.procs.2021.08.057
[28] https://doi.org/10.24963/ijcai.2020/367
[29] https://doi.org/10.1145/3328485
[30] https://doi.org/10.1145/3448248
[31] https://arxiv.org/pdf/2307.06435

10-MIN BREAK

PART 2:
AI AUDIT

Why do we need to audit AI

- AI is being built, adopted and deployed at an exponential rate.

- Widespread concerns about whether it/the companies developing it can be trusted because **unmanaged AI risks** are **threatening society** and **causing major financial and reputational damage**.

Twitter taught Microsoft's AI chatbot to be racist in less than a day [31]

A factual error by Bard AI chatbot just cost Google $100 billion [32]

Amazon scrapped sexist AI tool [33]

Face recognition police tools 'staggeringly inaccurate' [34]

Algorithm Audits

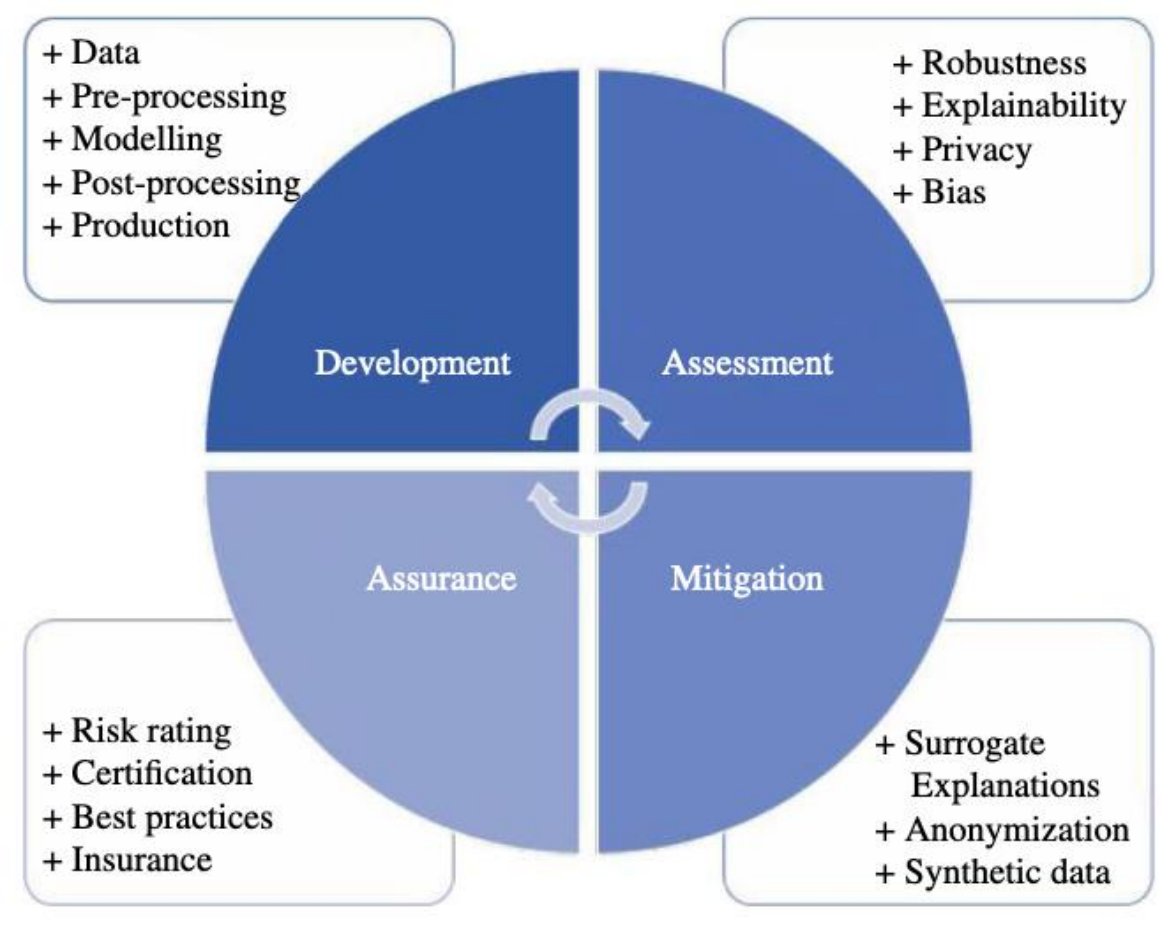
- A process of **independent and impartial system evaluation**, whereby an external auditor assesses an **AI system's efficacy, risk levels, and vulnerabilities** and **offers recommendations** to **mitigate risks to an acceptable level** [35].
- A new industry to professionalise and industrialise AI (incl. ML and associated algorithms) from legal, ethical and safety standpoints.

- Types of algorithm audits:
  - ❑ **Model Audits**: Assess the results generated by an Algorithm for Efficacy, Robustness, Fairness, Transparency and Privacy.
  - ❑ **Governance Audits**: Assess an AI developer's organisational procedures, accountability structures, oversight mechanisms and quality management systems.

AI Regulations Worldwide

Source – Holistic AI [36]

Systematising Algorithm Audits



[35]

Key Components of Algorithm Auditing – 1

- **Object of audit**: algorithms (data + model + development process + documentation).
- **What to audit / verticals of auditing**: development stages vs risk verticals – which ones to prioritise?

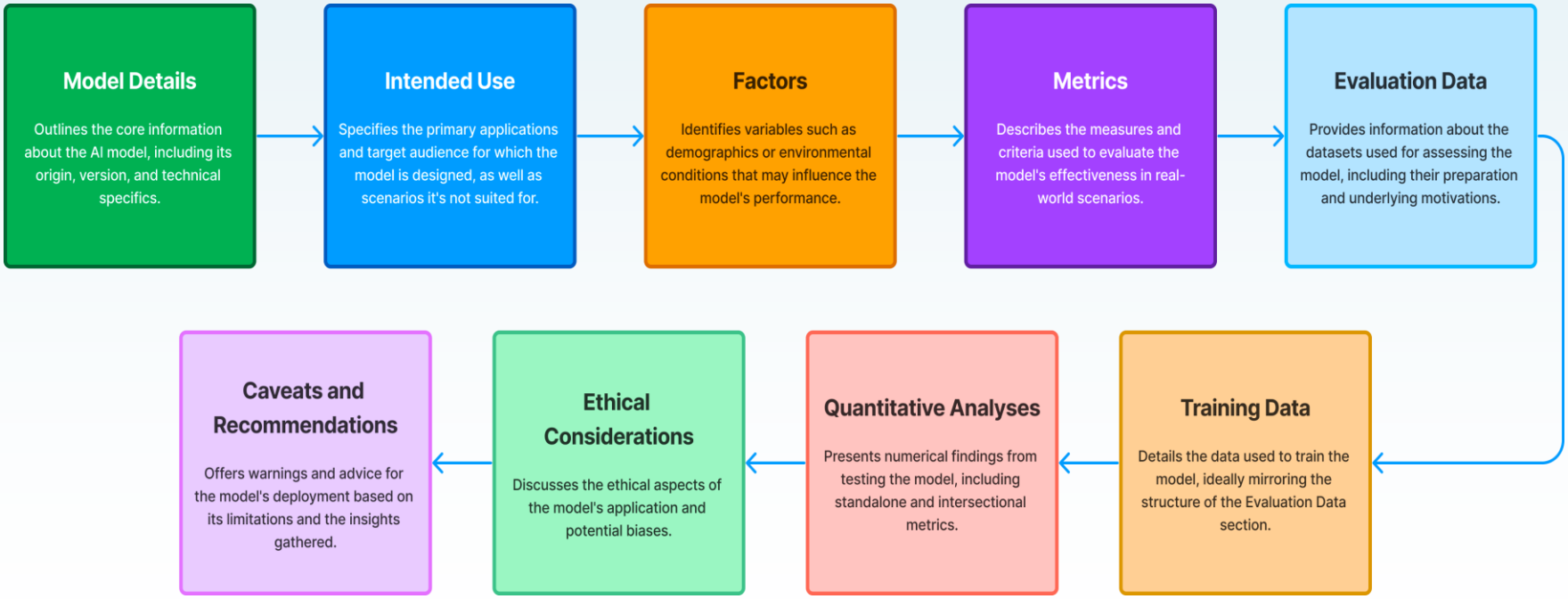
stage	explainability	robustness	fairness/bias	privacy
data and task setup	data collection and labelling	data accuracy	population balance	DPIA
feature pre-processing	dictionary of variables	feature engineering	fair representations	data minimization
model selection	model complexity	model validation	fairness constraints	differential privacy
post-processing and reporting	auxiliary tools	adversarial testing	bias metric assessment	model inversion
productionizing and deploying	interface and documentation	concept drift detection and continuous integration	real-time monitoring of bias metrics	rate-limiting and user's queries management

[35]

Ingredients of the Development Process

Documentation created as part of the algorithm-building phases are essential for (internal/external) risk mapping. Two of which are:

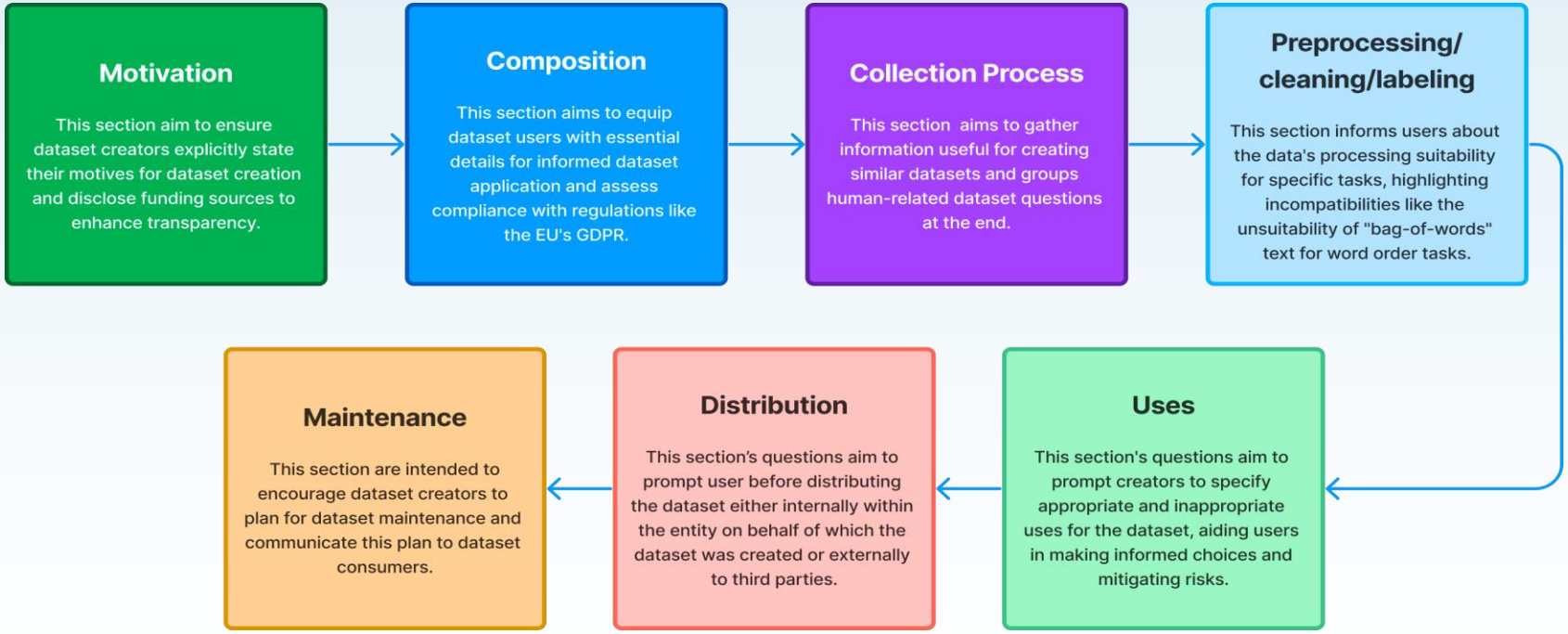
- Model Cards



Source – Holistic AI & Zekun Wu [36]

Ingredients of the Development Process

- Data Sheets



Source – Holistic AI
& Zekun Wu [36]

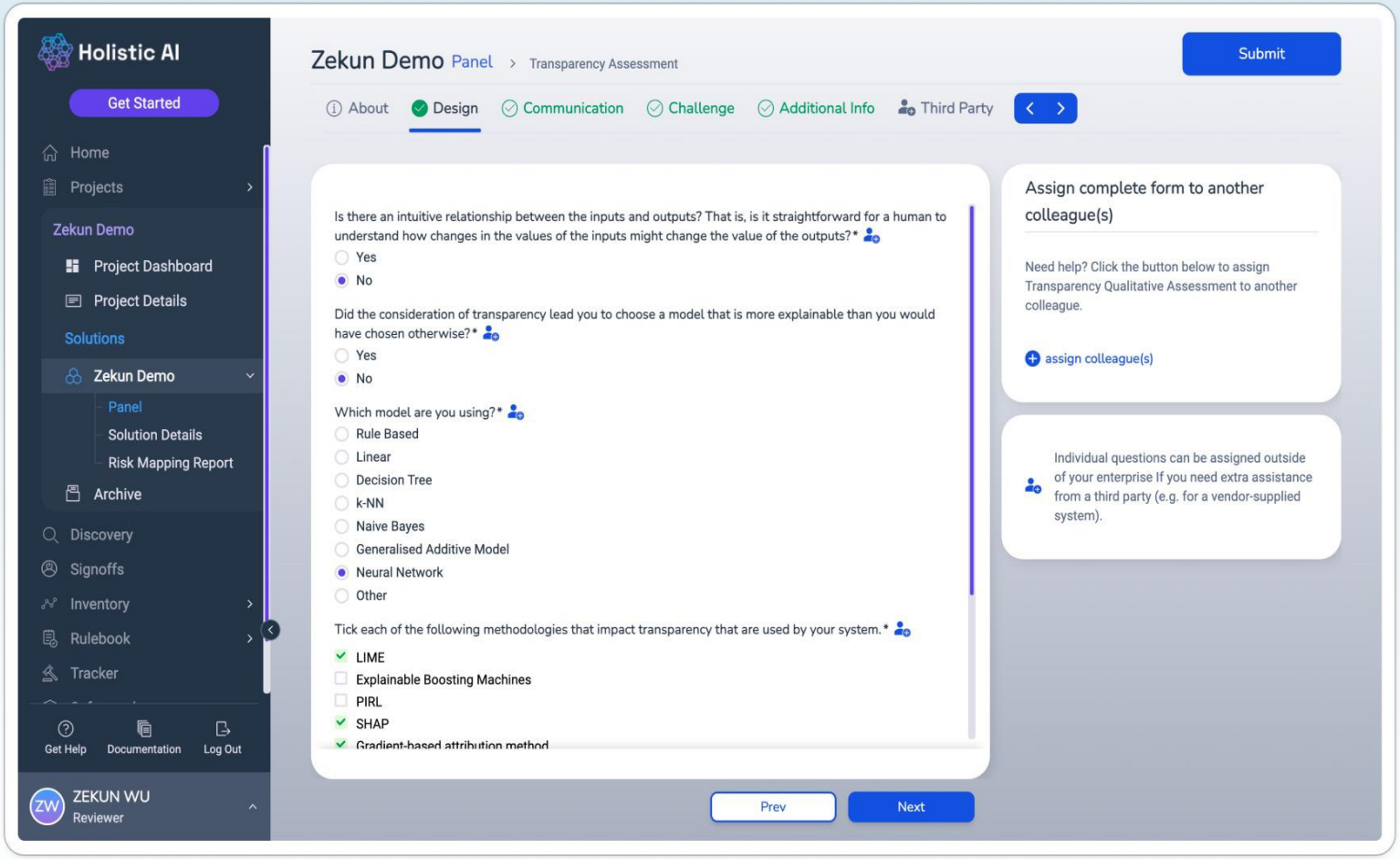
Key Components of Algorithm Auditing – 2

- **Ways to audit / levels of access:** more granular than ”white-box” and “black-box” based on knowledge/access to the components behind the learning process.

dimension	level 1 process access	level 2 model access (.)	level 3 input access $f(x)$	level 4 outcome access $f(x), y$	level 5 parameter control $f_\theta(x), y$	level 6 learning goal $L(f_\theta(x), y)$	level 7 ‘white-box’
explainability	checklist	feature relevance partial dependency	surrogate explanations	accuracy of explanations	stability of explanations	model complexity	documents and specific explanations
robustness	checklist	adversarial attacks	synthetic data	concept drift analysis	stability analysis	stress-testing	model selection and validation
fairness	checklist	adversarial fairness	bias in outcome	bias in opportunity	stability of bias metrics	trade-off of bias and loss metric	model selection and development
privacy	checklist	statistical disclosure	property and membership inference	inversion attacks	functionality stealing	model extraction	model security evaluation
information concealed	very high	high	high	high/medium	medium	medium	low
feedback detail	low	medium	medium	high/medium	high	high	very high
typical application	sales forecasting	cyber security	recruitment	credit-scoring	facial recognition	algorithmic trading	self-driving vehicle
appropriate oversight	guidelines	external auditing/ certification	external auditing	external auditing	external auditing	internal/external auditing	internal auditing

[35]

Qualitative Assessments at Holistic AI



Tech Stack

Recording of Tech details: Model Card/ Dataset Sheet

4

Verification Qualitative Testing

5

Assessment & Mitigation
Safeguard & Governance Platform

Source – Holistic AI & Zekun Wu [36]

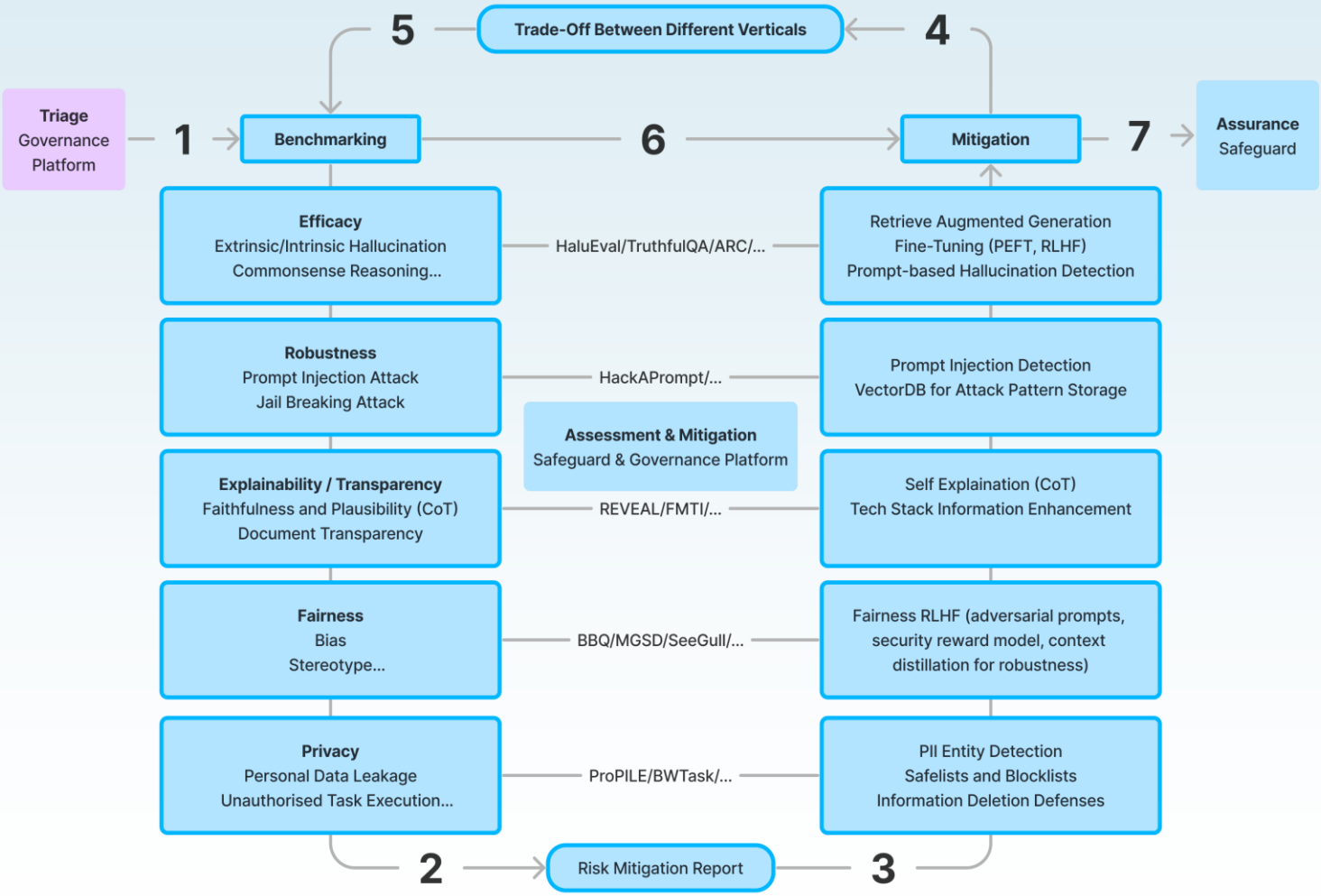
Key Components of Algorithm Auditing – 3

- **After audit / mitigation strategies**: Audit feedback to improve algorithm is likely to be more targeted, technical, diverse and effective if there is greater level of access.

stage	explainability	robustness	fairness/bias	privacy
data and task setup	dictionary of variables and datasheets	collecting targeted data, reframing loss function	alternative data sources	anonymization
feature pre-processing	avoiding excessive feature engineering	feature squeezing	synthetic data	dimensionality reduction
model selection	by-design interpretable models	adversarial training	counterfactual fairness	federated learning
post-processing and reporting	LIME, SHAP	high confidence predictions and confidence intervals	calibrated odds	model inversion mitigation
productionizing and deploying	recourse interface	'circuit-breaking'	monitoring panels	rate-limiting and user's queries management

[35]

Assessment and Mitigation at Holistic AI



Source – Holistic AI
& Zekun Wu [36]

Quantitative Assessment – Benchmarking – 1

- A success-testing intervention, which assesses an AI system's performance against a **predefined task**, such as mapping AI system outputs to a dataset of prompts and responses [37, 38, 39].

- Can be deployed throughout model development for iteratively improving model efficacy, or "**hill-climbing**".

- Measures levels of **model efficacy** (aggregate benchmarks (MMLU), comprehension (HellaSwag), math capabilities (MATH), factuality (FACTOR), etc), and model harms (representational biases (BOLD, BBQ), mis/disinformation (FEVER), toxicity (ToxiGen) in generated text, etc).

- Used to compare **relative efficacy** with other models, or a model's different development iterations.

Source – Holistic AI
& Zekun Wu [36]

Quantitative Assessment – Benchmarking – 2

- **Lack of fit-for-purpose benchmarks** across model harms and modalities

- **Risks of Memorisation**, or the model inadvertently training on public benchmarks; reducing accuracy

- **Resource intensive**, requiring significant tooling support

- **Capability Overhang**: Over-reliance on benchmarks may lead to unknown risks being ignored [40].

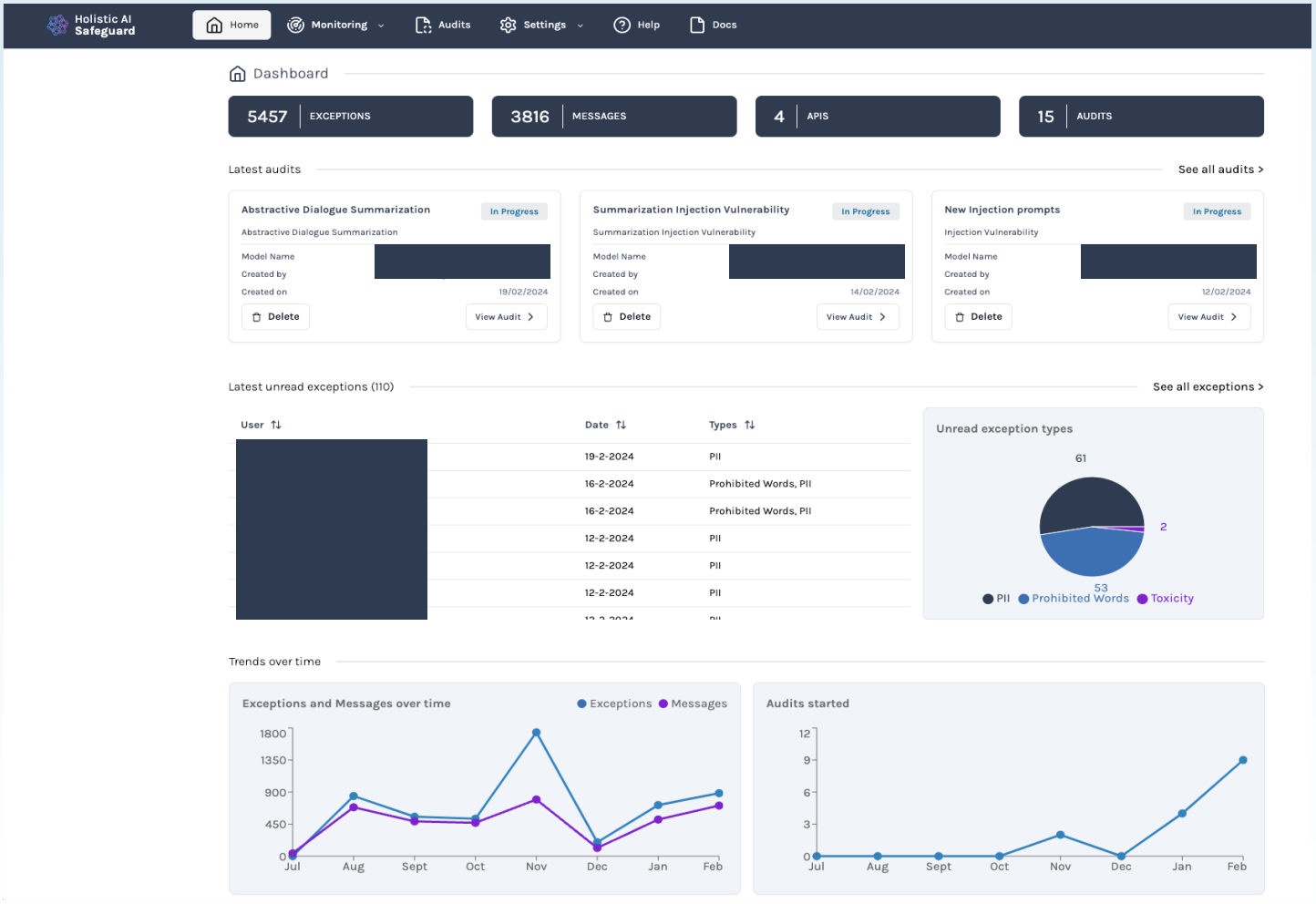
Source – Holistic AI
& Zekun Wu [36]

Key Components of Algorithm Auditing – 4

- **Outcome of audit / assurance processes**: After applying mitigation strategies, assess conformance to regulatory, governance and ethical standards → exhibit trustworthiness.
  - ❑ **General and sector-specific assurance**: national (US NIST, UK ICO, EU AI Act, etc.), sectoral (finance: SEC, FCA, etc.; health: NIH, NHS; etc.)
  - ❑ **Governance**: technical assessments (robustness, privacy, etc.), and impact assessments (risk, compliance, etc.).
  - ❑ **Unknown risks**: risk schemes and highlighting “red-teaming”.
  - ❑ **Monitoring interfaces**: risk assessments and ”traffic-light” interfaces.
  - ❑ **Certification**: of the system or the AI engineers themselves (i.e., training).

- A tangential point is that the regulators themselves may need to use AI, e.g., RegTech (Regulatory Technology) and SupTech (Supervisory Technology).

Ongoing Monitoring at Holistic AI



Model level- Model Onboarding - API

2

Ongoing monitoring

Source – Holistic AI & Zekun Wu [36]

Example of Data Assurance Rating – UK ICO

Colour code	Internal audit opinion	Definitions
	High assurance	There is a high level of assurance that processes and procedures are in place and are delivering data protection compliance. The audit has identified only limited scope for improvement in existing arrangements and as such it is not anticipated that significant further action is required to reduce the risk of non-compliance with data protection legislation.
	Reasonable assurance	There is a reasonable level of assurance that processes and procedures are in place and are delivering data protection compliance. The audit has identified some scope for improvement in existing arrangements to reduce the risk of non-compliance with data protection legislation.
	Limited assurance	There is a limited level of assurance that processes and procedures are in place and are delivering data protection compliance. The audit has identified considerable scope for improvement in existing arrangements to reduce the risk of non-compliance with data protection legislation.
	Very limited assurance	There is a very limited level of assurance that processes and procedures are in place and are delivering data protection compliance. The audit has identified a substantial risk that the objective of data protection compliance will not be achieved. Immediate action is required to improve the control environment.

[41]

EU AI Act

- **The EU AI Act is a comprehensive legislative framework proposed by the European Commission to regulate artificial intelligence (AI) systems within the European Union.**

- Its goal is to **establish an ecosystem of trust**, ensuring AI technologies are developed and deployed in a way that is safe, transparent, and respects fundamental human rights.

- The Act **categorizes AI systems based on their risk levels**—ranging from minimal to unacceptable risk—and sets out corresponding regulatory requirements to mitigate potential harms.

- **Envisioned to be enforced by 2026**, it aims to **lead global standards for AI regulation**, balancing the promotion of innovation with the protection of individuals and societal values.

[42]

Missing Link

- From part 1, the auditing and regulatory bodies **don't seem to provide equal importance usability and usefulness**, right?

- Where is the link between the protection of human rights/societal values that these different regulatory bodies are working so hard to ensure, and **providing accessibility and equal access in the first place**?

- Perhaps this will be the next order of business.

- For now, it remains in the hands of developers to build and deploy AI ***responsibly***.

References for Part 2

[31] https://www.bbc.co.uk/news/technology-35890188

[32] https://time.com/6254226/alphabet-google-bard-100-billion-ai-error/

[33] https://www.bbc.co.uk/news/technology-45809919

[34] https://www.bbc.co.uk/news/technology-44089161

[35] https://doi.org/10.1098/rsos.230859

[36] https://www.holisticai.com/

[37] https://arxiv.org/abs/2206.08325

[38] https://arxiv.org/abs/2307.06908

[39] https://www.anthropic.com/news/evaluating-ai-systems

[40] https://arxiv.org/pdf/2305.15324

[41] https://ico.org.uk/for-organisations/audits/

[42] https://www.europarl.europa.eu/doceo/document/A-9-2023-0188-AM-808-808_EN.pdf

ANY QUESTIONS